

Palliative Nephrology

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Palliative care is defined by the World Health Organization as “an approach that improves the health-related quality of life of patients and their families facing the problem associated with life-threatening illness, through the prevention and relief of suffering by means of early identification and impeccable assessment and treatment of pain and other problems, physical, psychosocial, and spiritual.”¹ The term *supportive care* is usually preferred by patients and families and is the term used by the Kidney Disease: Improving Global Outcomes (KDIGO) summary document on supportive care in kidney disease² and likewise in this chapter. As shown in Fig. 94.1, supportive care should begin at the time of diagnosis of advanced kidney disease. Fig. 94.2 shows how the different components can be used at different times.

PROGNOSIS

Informing patients accurately about prognosis is key to individualizing dialysis and associated supportive care. Patients with advanced kidney disease often have other comorbidities and complications of treatment; these and increasing age are causes of a high burden of morbidity affecting quality of life (QOL) and shortening length of life. Physicians, however, often are not good at estimating prognosis, with quotes in recent literature such as, “If you’re on dialysis you could last 10, 15, 20 years (male, 76 years old)” or “You will probably have 6 years on dialysis (male, 82 years old).”³ Patients often want to have conversations about life expectancy,⁴ and failure to have these discussions can result in high expectations and increased desire for aggressive treatment.⁵ Being aware of prognosis is a key part of shared decision making.⁶ At the individual level, the surprise question of “Would you be surprised if the patient died in the next 6 or 12 months?” has been shown to enhance prognostic prediction,⁷ although it has the limitation of being subjective and depending on clinician perception.

Prognostic tools have therefore been developed to enable clinicians to estimate the prognosis of patients. Factors that can be used to develop tools are summarized in Box 94.1; more detailed information can be found in an overview by Couchoud et al.⁶

Prognostic tools, however, have limitations because they provide an estimate of average survival for a population of similar risk but not for the individual. Before using a particular tool, it is important to know whether the patient concerned is similar to those used for the development of the tool in terms of things such as age, ethnicity, kidney replacement therapy (KRT), and clinical data available. The tool that is most readily accessible is the Analyzing Data, Recognizing Excellence and Optimizing Outcomes (ARO) score, which is based on simple clinical and laboratory data for prevalent European patients on hemodialysis⁷; it can be easily calculated on the ARO Risk Score website.⁸ For patients who have not begun KRT, a risk score for mortality in the first 90 days of dialysis has been developed for patients older than 75 years from the French Renal Epidemiology and Information

Network (REIN) registry.⁹ In routine clinical practice, these tools can inform discussions with patients and families.

The prognostic tools discussed herein are all specific for kidney disease and mostly for patients on dialysis. Clinical tools useful as prognostic indicators in the general population are also applicable in the presence of kidney disease. The concept of frailty is rapidly becoming one of the useful prognostic indicators for hospitalization, complications of procedures, and mortality in the general population.¹⁰ Clinically, frailty manifests as a composite of poor physical function, exhaustion, low physical activity, and weight loss and is associated with an increased risk for falls, cognitive impairment, hospitalization, and death. Frailty is more common in the chronic kidney disease (CKD) population and, not surprisingly, is a strong predictor of mortality.¹¹ There are various methods of assessing frailty, but the easiest to use clinically is the Clinical Frailty Scale (Table 94.1), which correlates with more complex assessment methods and predicts hospitalization and mortality.¹² Using this scale, prior work has shown that it is frailty, not dialysis modality (assisted peritoneal dialysis [PD] and hemodialysis [HD]), that is associated with symptom score, physical functioning, and better QOL.¹³ Routinely determining a frailty score in older patients approaching or on dialysis enhances prognostication both of survival and QOL. Frailty therefore is another factor that should inform discussions with patients and families.

COMMUNICATION AND SHARED DECISION MAKING

Good communication is key to enabling people to optimize QOL and achieve their goals at all stages of a long-term condition. Conversations must include awareness of prognosis and wishes for end-of-life planning. These latter conversations are commonly referred to as “advance care planning,” but conversations about prognosis and realistic outcomes of potential treatments and their limitations should ideally be routinely conducted with people at all stages of advanced kidney disease. Treatment decisions should be made *with* and not *for* the patient, using shared decision making. This is a process in which patients, when faced with an important choice about their health care, can review all the treatment options available to them and participate actively with their health-care professional in making that decision,¹⁴ thereby respecting patient autonomy.¹⁵ Shared decision making is embedded in the U.S. Renal Physicians Association guidelines on dialysis initiation and withdrawal,¹⁶ which state that “clinicians, family members, and others have an ethical duty to accept the decisions regarding medically indicated treatment made by competent patients and, in the absence of competence, to formulate decisions that would respect patients’ wishes, or if wishes are unknown, advance the best interest of their patients.” To make these decisions, patients and families must be given evidence-based information about what is happening regarding disease process, what is likely to

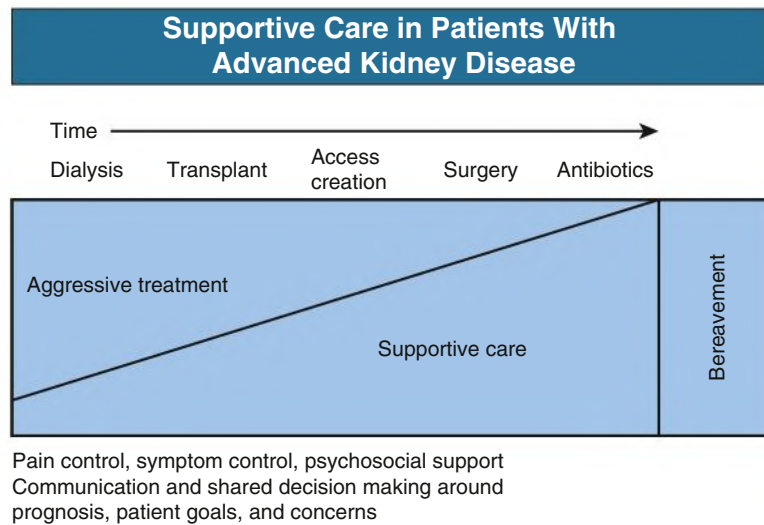


Fig. 94.1 Schematic diagram of how supportive care fits into patient pathway with advanced kidney disease.

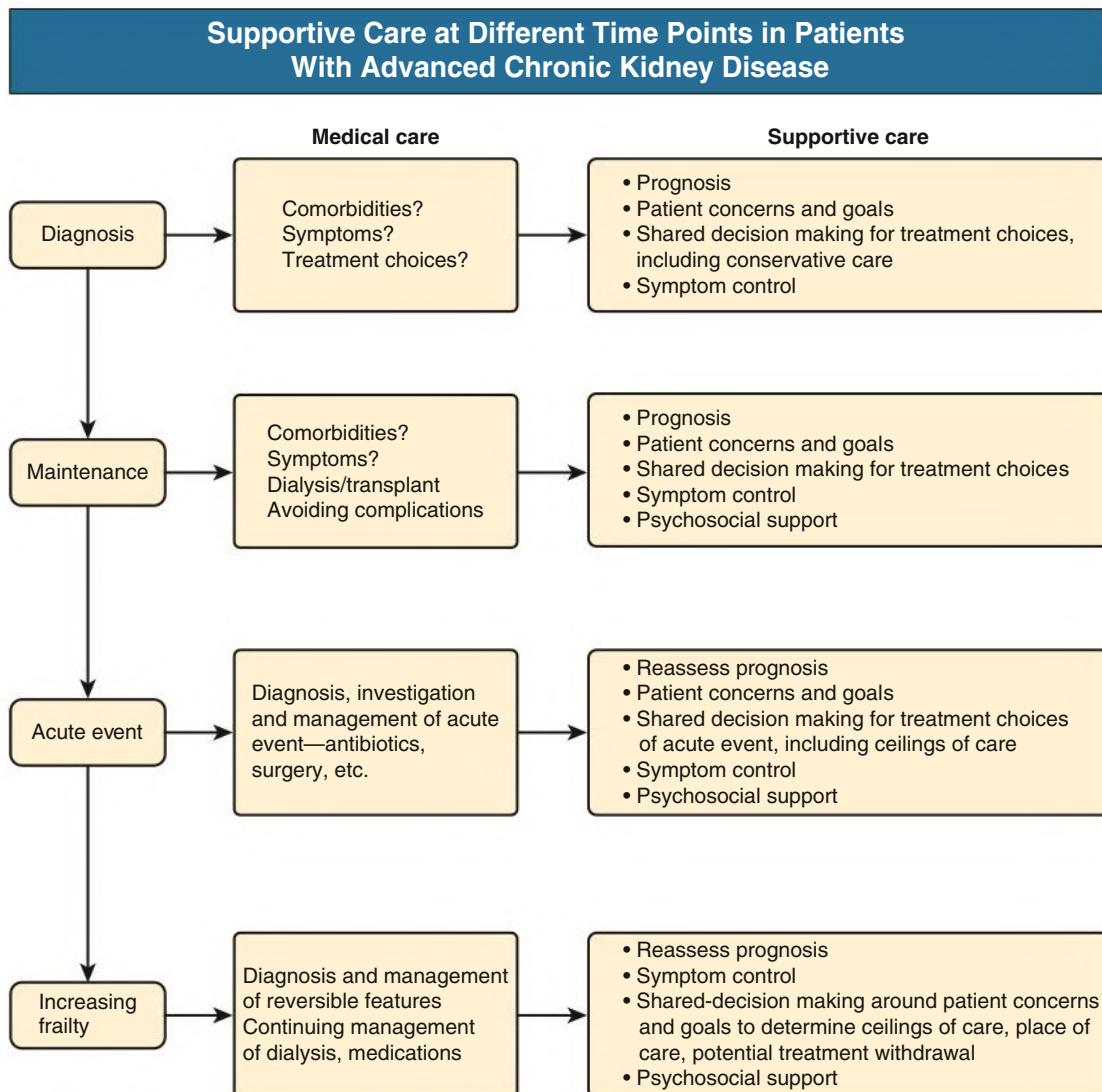


Fig. 94.2 Use of supportive care at different time points in life with advanced kidney disease.

BOX 94.1 Factors Used to Develop Prognostic Tools

Clinical Data

- Age
- Assistance with activities of daily living
- Sex
- Congestive heart failure
- Peripheral vascular disease
- Dysrhythmia
- Active malignancy
- Mobility
- Diabetes
- Body mass index
- Behavioral disorders
- Frequency of hospitalization

Dialysis Related

- Use of central vein dialysis catheter
- Early nephrology referral

Laboratory Values

- Plasma albumin
- Plasma creatinine
- C-reactive protein

BOX 94.2 Time Points for Advance Care Planning Discussions

- As part of kidney replacement therapy planning, particularly if transplant ineligibility or conservative care is discussed.
- Starting or within first few months of starting dialysis.
- Not eligible for transplant and patient does not want dialysis or is anticipated to have poor outcomes on dialysis.
- Not responding to peritoneal dialysis (PD) and patient does not want hemodialysis (HD) or is anticipated to have poor outcomes on HD and transplantation is not feasible.
- Recurrent vascular access problems in HD patient who does not want or cannot transfer to PD, and transplantation is not feasible.
- Patient wants to withdraw from dialysis.
- After intercurrent event that results in worsening of prognosis, such as stroke, fall and bone fracture, worsening cognitive function, or new malignancy.
- Development of intercurrent event needing major surgical intervention, such as coronary artery bypass surgery or major abdominal surgery.
- Increasing frailty and/or cognitive decline.
- Answer of “no” to surprise question: “Would you be surprised if patient died in the next 12 months?”

TABLE 94.1 Clinical Frailty Scale

Score	Definition
1	<i>Very fit:</i> Robust, active, energetic, well-motivated and fit
2	<i>Well:</i> No active disease but less fit than people in category 1
3	<i>Well, with treated comorbid disease:</i> Disease symptoms are well controlled compared with category 4
4	<i>Apparently vulnerable:</i> Although not frankly dependent, commonly complain of being “slowed up”
5	<i>Mildly frail:</i> With limited dependence on others for instrumental activities of daily living
6	<i>Moderately frail:</i> Help is needed with both instrumental and noninstrumental activities of daily living
7	<i>Severely frail:</i> Completely dependent on others for the activities of daily living or terminally ill

From Clegg A, Young J, Iliffe S, et al. Frailty in elderly people. *Lancet*. 2013;381:752–762.

happen regarding prognosis, what treatment options are available, and what the realistic outcomes are from these treatments. Healthcare professionals should seek to understand the patient’s concerns and lifestyle priorities and then use these to guide the patient through decision making about, for example, wishes regarding specific treatments, ceilings/limits of care if they are no longer able to make decisions, and place of care at end of life. A summary of when such conversations could and should take place is shown in [Box 94.2](#).

There are challenges to having advance care planning discussions, such as time needed, cultural and linguistic challenges,¹⁷ and overcoming the inhibitions of healthcare professionals to have difficult conversations, particularly when the prognostic future is uncertain.¹⁸ Some useful questions to help with overcoming these challenges are shown in [Box 94.3](#).

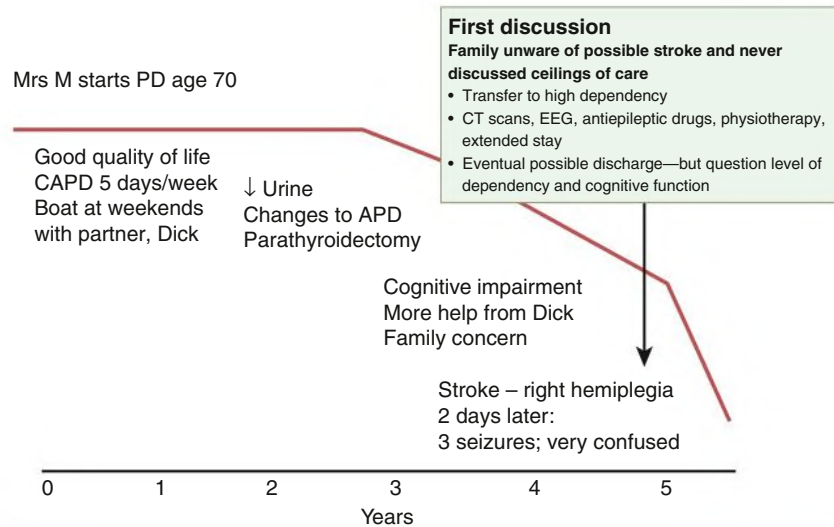
Conversations about prognosis and advance care planning should be embedded in nephrologic practice. The timing and frequency of discussions can have a huge influence on patient outcomes. [Fig. 94.3](#)

BOX 94.3 Questions and Statements to Use in Advance Care Planning

- How is your health now compared with a few months ago?
- Would you be surprised if I told you that things are not going well?
- It is always hard to predict the future, but it would not be a surprise if some major acute event occurred or your health deteriorated significantly in the next few months (or other applicable time scale). We should therefore discuss what your wishes would be should this occur.
- It is a good idea to talk about what we can do and how to get help if you proceed to become more unwell.
- Different people make decisions in different ways. Do you make decisions on your own, do you involve your family, or does someone from your family make important decisions for you?
- Do you have a faith or spiritual beliefs that help you in difficult times?
- Have you ever thought how much treatment you would want if you are very ill, unable to communicate to make your own decisions, and unlikely to leave hospital and be independent?
- You know you have a choice about where you would like your end-of-life care to take place. Would you like to talk through the options?
- Some people decide to stop dialysis when their health has deteriorated and they feel that dialysis has become a burden and is no longer of benefit. We do not need to mention that any further now, but it is something you may want to discuss later.

gives the example of a woman who has had a major stroke 5 years after starting dialysis at age 70 years and who has had progressive cognitive impairment in the last 2 years. In the absence of any previous discussions, the family would be unaware of the high probability of stroke and, in all likelihood, would not have had discussions about extent of treatment in such an event; there would then be a high probability of them expecting intensive investigations and management. In reality, for this particular patient, there had been a number of discussions after the onset of cognitive impairment and the patient had made it clear that she did not want continuing active management if she was not going to be independent; this made the final decision making by the family much easier. This has been confirmed by randomized controlled

Scenario 1: First Discussion With Family at Time of Major Stroke 5 Years After Starting Dialysis



Scenario 2: Discussions Throughout Timespan on Dialysis

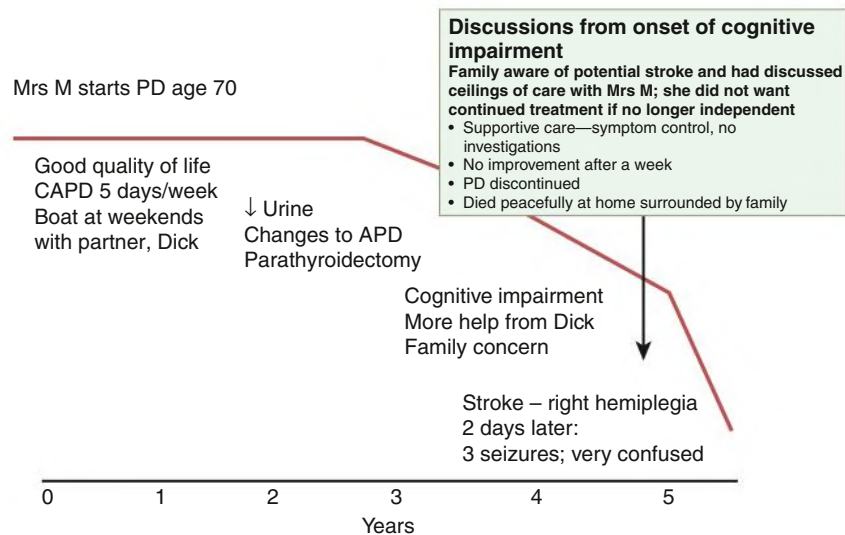


Fig. 94.3 How Communication and Advance Care Planning Can Influence Patient Outcomes. *Scenario 1:* First discussion with family at time of major stroke 5 years after starting dialysis. *Scenario 2:* Discussions throughout time span on dialysis. APD, Ambulatory peritoneal dialysis; CAPD, continuous ambulatory peritoneal dialysis; CT, computed tomography; EEG, electroencephalogram; PD, peritoneal dialysis.

studies in elderly and dialysis populations showing better QOL at end of life and improved certainty about decision making by close family members when these conversations have occurred.^{19,20}

CONSERVATIVE CARE

Conservative (nondialytic) kidney care has been defined (Box 94.4), using the term *comprehensive conservative care* and encompassing all the components required for effective comprehensive care.

It has also been proposed that globally there are three distinct groups within the conservative care population (Box 94.5); this reflects limited availability of KRT, the consequent (lack of) options for choice

in low-income and middle-income countries, and constraints on diagnosis of stage 5 CKD in resource-poor settings.

Rates of conservative management vary considerably among and within countries, and it is likely that individual nephrology practice is a factor in this variation.²¹ It is important therefore to consider all available evidence on survival of those managed conservatively and to contrast this to evidence on survival with dialysis. However, this is challenging, because those who are more fit usually opt for dialysis, and others are advised to follow conservative nondialytic management because of comorbidity or other factors that adversely influence survival. This selection bias makes it hard to meaningfully compare survival between those with dialysis and those managed conservatively.

BOX 94.4 Definition of Comprehensive Conservative Care

Comprehensive conservative care is planned, holistic, patient-centered care for patients with stage 5 (glomerular filtration rate category 5) chronic kidney disease that includes:

- Interventions to delay progression of kidney disease and minimize risk for adverse events or complications
- Shared decision making
- Active symptom management
- Detailed communication, including advance care planning
- Psychological support
- Social and family support
- Cultural and spiritual domains of care

Comprehensive conservative care does not include dialysis.

From Davison SN, Levin A, Moss AH, et al. Executive summary of the KDIGO Controversies Conference on Supportive Care in Chronic Kidney Disease: Developing a roadmap to improving quality care. *Kidney Int.* 2015;88:447–459.

BOX 94.5 Distinct Conservative Care Populations

Comprehensive conservative care: Conservative care that is chosen or medically advised.

Choice-restricted conservative care: Conservative care for patients in whom resource constraints prevent or limit access to kidney replacement therapy; therefore a choice for conservative care cannot be recognized.

Unrecognized stage 5 chronic kidney disease: Chronic kidney disease is present but has not been recognized or diagnosed; therefore a choice for conservative care cannot be recognized.

From Davison SN, Levin A, Moss AH, et al. Executive summary of the KDIGO Controversies Conference on Supportive Care in Chronic Kidney Disease: Developing a roadmap to improving quality care. *Kidney Int.* 2015;88:447–459.

The summary of the evidence by O'Connor and Kumar²² provides a good overview of current evidence, although further comparative studies^{23–27} have since been published. This collected evidence shows that there is a substantial survival advantage with dialysis; however, for older people with end-stage kidney disease (ESKD; >75–80 years) with comorbidity, much of this survival advantage is lost. When time in hospital and receiving dialysis is also considered, there is little or no survival advantage to be gained from dialysis for these older patients with comorbidities. However, there are two notes of caution in interpreting and applying this evidence; first, the evidence from those receiving conservative care remains very limited, and second, the conservative care population is heterogeneous. This makes it much harder to relate to individual circumstances, yet decisions for conservative care always need to be individually tailored.

It is important to also consider symptoms, QOL, illness experience, and survival. There is robust evidence that patients rarely prioritize survival above all else; QOL is equally important and often takes precedence.²⁸ The systematic review cited previously²⁹ shows that patients managed conservatively (as do those receiving dialysis) report significant symptom burden, with an average of 9 to 17 symptoms. QOL is generally similar between patients on dialysis and patients receiving conservative care, although there may be some decrease in life satisfaction in relation to dialysis adjustment.³⁰ Of note, with good supportive care, many of the conservatively managed patients have stable or improved symptoms and QOL.³¹

BOX 94.6 Situations Where Dialysis Withdrawal Is Appropriate

- Patients with decision-making capacity who, being fully informed and making voluntary choices, refuse dialysis or request that dialysis be discontinued
- Patients who no longer possess decision-making capacity
 - Those who have previously indicated refusal of dialysis
 - Those whose legal agents/surrogates request that it be discontinued
- Patients with irreversible, profound neurologic impairment such that they lack signs of thought, sensation, purposeful behavior, and awareness of self and environment

From Davison SN, Levin A, Moss AH, et al. Executive summary of the KDIGO Controversies Conference on Supportive Care in Chronic Kidney Disease: Developing a roadmap to improving quality care. *Kidney Int.* 2015;88:447–459.

DIALYSIS WITHDRAWAL

As already demonstrated in the case discussed in Fig. 94.3, patients may decide to stop dialysis. The median time to death after stopping dialysis is 8 to 9 days³² but can be considerably longer if there is significant residual kidney function. There is information for patients and caregivers about stopping dialysis on both U.K. and U.S. patient support websites.^{33,34} The situations in which dialysis withdrawal is appropriate are shown in Box 94.6.

Studies suggest that the rate of dialysis withdrawal has increased over the years, but this needs to be interpreted with caution because there is no well-accepted definition of dialysis withdrawal.³⁵ A recent report published from the Scottish Renal Registry³⁶ has defined death resulting from withdrawal as patients refusing further treatment, therapy ceased for other reasons, and dialysis withdrawn for medical reasons. In this study, dialysis withdrawal was the primary cause of death in 19% of all deaths (a rate of 41 in 1000 patient-years) over a 7-year period; in a further 17% of patients, withdrawal from dialysis was considered to have contributed to death but not to have been the main cause. Using this definition, deaths attributed to dialysis withdrawal accounted for around one-third of deaths; this did not change over the 7 years of the study but did vary significantly across the different kidney centers with dialysis withdrawal accounting for 3.3% to 55.8% of all deaths per center.³⁶ This variation also has been found on a larger scale between different European countries, with higher reports of dialysis withdrawal if respondents in the survey worked in a public center, if stopping life-supporting treatments was perceived as allowed, if withdrawal decisions were considered shared between patients and doctors, and if palliative care was reimbursed by payers.³⁷ The patient factors associated with dialysis withdrawal vary in individual studies, but many of the following have been consistently identified: older age, female sex, higher comorbidity burden (particularly cerebrovascular disease), being on HD, early initiation of dialysis, and late referral to nephrologist.^{36,38,39} Cultural differences also play an important role in dialysis withdrawal with lower rates of dialysis discontinuation in racial and ethnic minorities reported in studies in English-speaking countries.^{38,39,40}

Withdrawal from dialysis is ethically and clinically acceptable only after a process of shared decision making with the patient (and close persons if the patient is no longer able to make decisions). These discussions need to be attuned to the culture (ethnic and religious) of the individual.¹⁷ It is also incumbent on all providers caring for a patient who is contemplating stopping dialysis to address potentially remedial factors contributing to the decision, such as depression, or other

symptoms, such as pain and potentially reversible social factors.² *It is essential to ensure access to appropriate supportive and/or hospice care as an integral part of care after any decision to withdraw dialysis.*² Details of this are discussed in the following section.

SYMPTOM CONTROL AND MANAGEMENT OF LAST DAYS

Symptoms are as common in those managed conservatively as among those on dialysis, with 3 in 4 reporting weakness and pruritus; 1 in 3 reporting drowsiness, breathlessness, and edema; and more than 1 in 2 patients with ESKD reporting pain.⁴¹ These symptoms are not mild; a high proportion (more than two-thirds) of those with pain, for instance, have moderate or severe pain, often related to comorbid conditions rather than to kidney disease. In addition, symptoms often accumulate, with an average of 12 symptoms experienced by each patient that impair QOL and need optimal management.

Evidence on management of symptoms is often extrapolated from other populations or based on best understanding of the

pharmacodynamics and pharmacokinetics in severe kidney impairment. This evidence is limited but is well summarized in a recent series of articles.⁴²

In the last days of life, additional symptoms may arise from uremia (nausea, vomiting, drowsiness, myoclonus, pruritus), fluid overload (breathlessness and edema), and immobility and poor tissue perfusion (pain, muscle soreness, pressure sores). It is best practice to prescribe “as required” doses of an analgesic, an antiemetic, a sedative, and an antisecretory agent for all patients with ESKD as they enter the last few days of life, to anticipate any distressing symptoms and ensure they can be rapidly addressed. The actual drugs used will vary between different centers and countries. What is important is involvement with local palliative care team support in determining which drugs to prescribe for the individual patient.

As end of life approaches, both patient and family often need much more information and explanation than is provided by health professionals. Encouragement to ask questions, straightforward but sensitively judged responses, and frequent opportunities for professionals to check on concerns and anxieties can go a long way to provide support to the patient and family at a difficult and challenging time.

SELF-ASSESSMENT QUESTIONS

- Which of the following factors have been shown to be associated with a poor survival prognosis in an 80-year-old man who has been on dialysis for 2 years?
 - Being on peritoneal dialysis
 - Peripheral vascular disease
 - Use of central venous catheter
 - Requiring assistance to get dressed
 - Low plasma albumin
- Advance care planning should include which of the following?
 - Legal process resulting in advance decision
 - Informing the patient that they will not be resuscitated
 - Discussion about prognosis
 - Identification of patient concerns
 - Opportunity for patient to record future wishes
- Which of the following statements can be made about dialysis withdrawal?
 - Approximately 20% of deaths of dialysis patients are stated to be because of dialysis withdrawal.
 - Death invariably occurs within 3 days of stopping dialysis.
 - Dialysis withdrawal can be viewed as euthanasia.
 - Shortness of breath is the main symptom after withdrawing dialysis.
 - Dialysis can be stopped in patients lacking decision-making capacity who have made a previous decision to this effect.
- Which of the following statements can be made about conservative care?
 - Survival rates of multimorbid patients older than 80 years are similar on dialysis and conservative care.
 - Survival of functionally independent patients 70 to 75 years old are the same on dialysis and conservative care.
 - Patients choosing conservative care should be persuaded to start dialysis if they deteriorate acutely.
 - Anemia management with erythropoietin is appropriate for patients choosing conservative care.
 - Conservative care is not “rationing” because patients choose not to have dialysis themselves.

REFERENCES

1. WHO Definition of Palliative Care. World Health Organization. <http://www.who.int/cancer/palliative/definition/en/>.
2. Davison SN, Levin A, Moss AH, et al. Executive summary of the KDIGO Controversies Conference on supportive care in chronic kidney disease: developing a roadmap to improving quality care. *Kidney Int.* 2015;88:447–459.
3. Tonkin-Crine S, Okamoto I, Laydon GM, et al. Understanding by older patients of dialysis and conservative management for chronic kidney failure. *Am J Kid Dis.* 2015;65:443–450.
4. Davison SN. End-of-life care preferences and needs: perceptions of patients with chronic kidney disease. *Clin J Am Soc Nephrol.* 2010;5:195–204.
5. Wachterman MW, Marcantonio ER, Davis RB, et al. Relationship between the prognostic expectations of seriously ill patients undergoing hemodialysis and their nephrologists. *JAMA Intern Med.* 2013;173:1206–1214.
6. Couchoud C, Hemmelgam B, Kotanko P, et al. Supportive Care: time to change our prognostic tools and their use in CKD. *Clin J Am Soc Nephrol.* 2016. <https://doi.org/10.2215/CJN.12631115>.
7. Floege J, Gillespie IA, Kronenberg F, et al. Development and validation of a predictive mortality risk score from a European hemodialysis cohort. *Kidney Int.* 2015;87:996–1008.
8. Risk ARO Score Calculators. <http://aro-score.askimed.com/>.
9. Couchoud CG, Beuscart J-P, Aldigier J-C, et al. On behalf of the REIN registry. Development of a risk stratification algorithm to improve patient-centered care and decision making for incident elderly patients with end-stage renal disease. *Kidney Int.* 2015;88:1178–1186.
10. Clegg A, Young J, Iliff S, et al. Frailty in elderly people. *Lancet.* 2013;381:752–762.
11. Alfaadhel TA, Soroka SD, Kiberd BA, et al. Frailty and mortality in dialysis: evaluation of a clinical frailty scale. *Clin J Am Soc Nephrol.* 2015;10:832–840.
12. Rockwood K, Song X, MacKnight C, et al. A global clinical measure of fitness and frailty in elderly people. *Can Med Assoc J.* 2005;173:489–495.
13. Iyasere OU, Brown EA, Johansson L, et al. Quality of life and physical function in older patients on dialysis: a comparison of assisted peritoneal dialysis with hemodialysis. *Clin J Am Soc Nephrol.* 2016;11:423–430.
14. NHS England. Shared decision making. <https://www.england.nhs.uk/ourwork/pe/sdm/#>.
15. Muthalagappan S, Johansson L, Kong WM, Brown EA. Dialysis or conservative care for frail older patients: ethics of shared-decision making. *Nephrol Dial Transplant.* 2013;28:2717–2722.
16. Renal Physicians Association. *Shared Decision-Making in the Appropriate Initiation of and Withdrawal from Dialysis (Clinical Practice Guideline)*. 2nd ed. Rockville: MD; 2010.
17. Brown EA, Bekker HL, Davison SN, et al. Supportive care: communication strategies to improve cultural competence in shared decision making. *Clin J Am Soc Nephrol.* 2016;11(10):1902–1908.
18. Kimbell B, Murray SA, Macpherson S, Boyd K. Embracing inherent uncertainty in advanced illness. *BMJ.* 2016;354:i3802.
19. Detering KM, Hancock AD, Reade MC, Silvester W. The impact of advance care planning on end of life care in elderly patients: randomised controlled trial. *BMJ.* 2010;340:c1345.
20. Song M-KS, Ward SE, Fine JP, et al. Advanced care planning and end-of-life decision making in dialysis: a randomised controlled trial targeting patients and their surrogates. *Am J Kid Dis.* 2015;66:813–822.
21. Kee F, Patterson CC, Wilson EA, et al. Stewardship or clinical freedom? variations in dialysis decision making. *Nephrol Dial Transplant.* 2000;15(10):1647–1657.
22. O'Connor NR, Kumar P. Conservative management of end-stage renal disease without dialysis: a systematic review. *J Pall Med.* 2012;15(2):228–235.
23. Hussain JA, Mooney A, Russon L. Comparison of survival analysis and palliative care involvement in patients aged over 70 years choosing conservative management or renal replacement therapy in advanced chronic kidney disease. *Pall Med.* 2013;27(9):829–839.
24. Da Silva-Gane M, Wellsted D, Greenshields H, et al. Quality of life and survival in patients with advanced kidney failure managed conservatively or by dialysis. *Clin J Am Soc Nephrol: CJASN.* 2012;7(12):2002–2009.
25. Kwok WH, Yong SP, Kwok OL. Outcomes in elderly patients with end-stage renal disease: comparison of renal replacement therapy and conservative management. *Hong Kong J Nephrol.* 2016;19:42–56.
26. Martínez Echevers Y, Toapanta Gaibor NG, Nava Pérez N, et al. Survival of patients ≥70 years with advanced chronic kidney disease: dialysis vs. conservative care. *Nefrologia.* 2016;36(3):283–291.
27. Chandna SM, Carpenter L, Da Silva-Gane M, et al. Rate of decline of kidney function, modality choice, and survival in elderly patients with advanced kidney disease. *Nephron.* 2016;134(2):64–72.
28. Higginson IJ, Gomes B, Calanzani N, et al. Priorities for treatment, care and information if faced with serious illness: a comparative population-based survey in seven European countries. *Palliat Med.* 2014;28(2):101–110.
29. O'Connor NR, Kumar P. Conservative management of end-stage renal disease without dialysis: a systematic review. *J Pall Med.* 2012;15(2):228–235.
30. Da Silva-Gane M, Wellsted D, Greenshields H, et al. Quality of life and survival in patients with advanced kidney failure managed conservatively or by dialysis. *Clin J Am Soc Nephrol: CJASN.* 2012;7(12):2002–2009.
31. Brown MA, Collett GK, Josland EA, et al. CKD in elderly patients managed without dialysis: survival, symptoms, and quality of life. *Clin J Am Soc Nephrol.* 2015;10(2):260–268.
32. Cohen LM, McCue JD, Germain MJ, Kjellstrand CM. Dialysis discontinuation: a 'good' death? *Arch Int Med.* 1995;155:42–47.
33. Kidney Care UK Saying no to dialysis. <https://www.kidneycareuk.org/about-kidney-health/order-or-download-booklets/>.
34. DaVita. What happens if someone stops dialysis? <https://www.davita.com/kidney-disease/dialysis/treatment/what-happens-if-someone-stops-dialysis?e/1521>.
35. Murphy E, Germain MJ, Cairns H, et al. International variation in classification of dialysis withdrawal: a systematic review. *Nephrol Dial Transplant.* 2014;29:625–635.
36. Findlay MD, Donaldson K, Doyle A, et al. On behalf of the Scottish Renal Registry (SRR). Factors influencing withdrawal from dialysis: a national registry study. *Nephrol Dial Transplant.* 2016;31:2041–2048.
37. van Biesen W, van de Luijngaarden MWM, Brown EA, et al. Nephrologists' perceptions regarding dialysis withdrawal and palliative care in Europe: lessons from a European Renal Best Practice survey. *Nephrol Dial Transplant.* 2015;30:1951–1958.
38. Ellwood AD, Jassal SV, Suri RS, et al. Early dialysis initiation and rates and timing of withdrawal from dialysis in Canada. *Clin J Am Soc Nephrol.* 2013;8:265–270.
39. Chan HW, Clayton PA, McDonald SP, et al. Risk factors for dialysis withdrawal: an analysis of the Australia and New Zealand Dialysis and Transplant (ANZDATA) Registry, 1999–2008. *Clin J Am Soc Nephrol.* 2012;7:775–781.
40. Agunbiade A, Dasgupta A, Ward Michael M. Racial/ethnic differences in dialysis discontinuation and survival after hospitalization for serious conditions among patients on maintenance dialysis. *J Am Soc Nephrol.* 2020;31:149–160.
41. Murtagh FE, Addington-Hall JM, Edmonds PM, et al. Symptoms in advanced renal disease: a cross-sectional survey of symptom prevalence in stage 5 chronic kidney disease managed without dialysis. *J Palliat Med.* 2007;10(6):1266–1276.
42. Davison SN, Tupala B, Wasylynyuk BA, Siu V, Sinnarajah A, Triscott J. Recommendations for the care of patients receiving conservative kidney management: focus on management of CKD and symptoms. *Clin J Am Soc Nephrol.* 2019;14:626–634.